Science practice booklet: Grade 9 expectations

Grade 9 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 9 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: HS-LS1-1, HS-LS1-3, HS-LS1-4, HS-LS1-5, HS-LS1-7, HS-LS3-1, HS-LS3-3

A worked example

One strand of a DNA molecule has the bases A T G G C A. Write the bases of the strand that pairs with it. Then say how many 3-base codons the original strand holds.

1. DNA bases pair in a fixed way: A pairs with T, and C pairs with G.
2. A T G G C A pairs with T A C C G T.
3. The original strand has 6 bases. Each codon is 3 bases, so 6 ÷ 3 = 2 codons.

The pairing strand is T A C C G T; the original strand holds 2 codons.

Questions

1. Which part of a cell is the main place where energy from food is released?

   A. Nucleus
   B. Mitochondria
   C. Cell wall
   D. Ribosome

2. Give two structures found in a plant cell but not in an animal cell, and say what each one does.

3. What is the job of the cell membrane? Why does a cell need it?

4. One DNA strand reads C G T A A G. Write the strand that pairs with it.

5. A sample of DNA has 30 percent adenine (A). What percentage of the bases is guanine (G)?

   A. 30 percent

   B. 40 percent

   C. 20 percent

   D. 60 percent

6. A gene is a section of DNA. What does a gene code for? If a gene region has 300 bases and every 3 bases code for one amino acid, how many amino acids are in the protein?

---

7. A scrape on a student's knee heals as skin cells divide by mitosis. When one skin cell divides, what results?

   A. Two cells, each with the same chromosomes as the original

   B. Two cells, each with half the chromosomes

   C. Four cells, each with half the chromosomes

   D. Four cells, each with the same chromosomes

8. A lab technician in Abu Dhabi watches one cell divide by mitosis, and each new cell divides again, for 4 rounds of division in total. How many cells are there at the end?

---

9. Give two reasons why the cells of a young date palm and of a growing child need to divide.

---

10. Which equation shows cellular respiration?

    A. Carbon dioxide + water + energy from light makes glucose + oxygen

    B. Glucose + carbon dioxide makes oxygen + water + energy released

    C. Oxygen + water makes glucose + carbon dioxide

    D. Glucose + oxygen makes carbon dioxide + water + energy released

11. In a school garden, tall (T) pea plants are dominant over short (t). Two tall plants, both with the genes Tt, are crossed. What fraction of their offspring are expected to be short?

    A. One quarter

    B. One half

    C. Three quarters

    D. None of them

12. On a hot day in the UAE the body sweats. Explain how this helps keep the body temperature steady, and name this kind of control.

Section 2: Earth and space

Standards practised: HS-ESS2-2, HS-ESS2-4, HS-ESS2-6, HS-ESS3-1, HS-ESS3-3

A worked example

A car gives off about 120 g of carbon dioxide for every kilometre it travels. A family drives 15 000 km in a year. How many kilograms of carbon dioxide does the car give off in that year?

1. Total in grams = 120 g × 15 000 = 1 800 000 g.
2. There are 1000 g in 1 kg, so divide by 1000: 1 800 000 ÷ 1000 = 1800 kg.
3. This carbon dioxide goes into the atmosphere, where it adds to the greenhouse effect.

1800 kg (1.8 tonnes) of carbon dioxide in a year.

Questions

1. Which pair matches an Earth system with a correct example from the UAE?

   A. Hydrosphere: the Hajar mountains

   B. Atmosphere: mangrove trees

   C. Biosphere: the water of the Arabian Gulf

   D. Geosphere: the sand dunes of the desert

2. Mangrove trees grow along the coast of the UAE. Describe how they connect three Earth systems.

---

3. Explain how the Sun's energy drives the water cycle.

---

4. On a UAE beach in the afternoon, the wind usually blows from the sea towards the land. What is the reason?

   A. Air over the sea rises because water heats faster

   B. Air over the land rises because land heats faster than water

   C. The Moon pulls air towards the land

   D. The Earth's spin pushes air onto the land

5. Which statement describes the climate of the UAE rather than the weather on one day?

   A. A dust storm reduced the view in Al Ain on Tuesday

   B. It rained in Fujairah yesterday evening

   C. Summers are usually hot and humid near the coast

   D. This morning was foggy in Abu Dhabi

6. Describe two ways carbon dioxide gets into the atmosphere and one way it is removed.

---

7. Explain the greenhouse effect in your own words. Why is it needed for life on Earth?

---

8. Why does burning more fossil fuels make Earth's average temperature rise?

   A. It releases extra carbon dioxide, which traps more heat

   B. It makes the Sun give off more heat

   C. It makes a hole in the ozone layer that lets in the Sun's heat

   D. It removes oxygen, and air with less oxygen holds more heat

9. A home uses 600 kWh of electricity in a 30 day month. Solar panels on its roof make 5 kWh each day. What percentage of the home's electricity do the panels supply?

---

10. Give one reason why solar energy is called renewable and coal is called non-renewable.

---

11. The Arabian Gulf takes in carbon dioxide from the air. What is one harmful result of this for coral reefs and shellfish?

    A. The water becomes saltier, so corals lose water

    B. The water becomes more acidic, so shells and skeletons are harder to build

    C. The water becomes less acidic, so shells grow too fast

    D. The water becomes clearer, so corals get too much light

12. Ice reflects much more sunlight than dark ocean water. Explain how melting sea ice can lead to even more warming.

Section 3: Matter, forces and energy

Standards practised: HS-PS1-1, HS-PS1-2, HS-PS1-5, HS-PS1-7

A worked example

Hydrogen gas burns in oxygen to make water. The unbalanced equation is H2 + O2 → H2O. Balance it.

1. Count the atoms on each side: left has 2 H and 2 O, right has 2 H and 1 O. The oxygen is not balanced.
2. Put 2 in front of H2O: now the right side has 4 H and 2 O.
3. Put 2 in front of H2: now the left side has 4 H and 2 O. Both sides match.

2 H2 + O2 → 2 H2O

Questions

1. Which statement about the particles in an atom is correct?

   A. Protons are positive and are found in the nucleus

   B. Neutrons are positive and are found in the nucleus

   C. Electrons are negative and are found in the nucleus

   D. Protons are negative and move around the nucleus

2. A sodium atom has atomic number 11 and mass number 23. How many protons, neutrons and electrons does the neutral atom have?

---

3. Chlorine has atomic number 17, so its electrons are arranged in shells as 2, 8, 7. How many valence electrons does a chlorine atom have?

   A. 8

   B. 17

   C. 7

   D. 2

4. Lithium, sodium and potassium are in the same group of the periodic table. Why do they react in similar ways?

---

5. Helium and neon are very unreactive gases. Use their outer electrons to explain why.

---

6. How do sodium and chlorine atoms form sodium chloride (table salt)?

   A. The two atoms share a pair of electrons

   B. Chlorine gives an electron to sodium

   C. Both atoms lose an electron

   D. Sodium gives an electron to chlorine, and the ions attract

7. A desalination plant in the UAE turns seawater into fresh water. In a water molecule (H_2O) the atoms share electrons. What is this kind of bond called, and how is it different from an ionic bond?

---

8. Which equation for the burning of magnesium is balanced?

   A. $Mg + O_2 \rightarrow 2\ MgO$

   B. $2\ Mg + 2\ O_2 \rightarrow 2\ MgO$

   C. $2\ Mg + O_2 \rightarrow 2\ MgO$

   D. $Mg + O_2 \rightarrow MgO_2$

9. A gas stove in a home burns methane (natural gas). Balance the equation: $CH_4 + O_2 \rightarrow CO_2 + H_2O$

---

10. 24 g of magnesium burns completely with 16 g of oxygen. What mass of magnesium oxide is made?

    A. 8 g

    B. 40 g

    C. 24 g

    D. 48 g

11. An iron nail left outside near the coast turns into rust, and the rust weighs more than the nail did. Has mass been created? Explain.

---

12. Photosynthesis is written as 6 CO2 + 6 H2O → C6H12O6 + 6 O2. Show that this equation is balanced by counting the atoms of each element on each side, and give one reason why changing the temperature or the amount of a reactant can change how fast a reaction goes.

Answer guide for parents

Section 1: Living things

1. B. Mitochondria. release energy from food in cellular respiration. If your child chose A, they mixed it up with the nucleus, which holds the DNA.
2. A plant cell has a cell wall (supports and protects), chloroplasts (make food by photosynthesis) and a large central vacuole (stores water). An animal cell has none of these, so any two with their jobs is correct.
3. The membrane is like a gatekeeper: it lets in nutrients and oxygen and lets wastes out. That keeps the inside of the cell different from the outside.
4. Pair each base: C with G, G with C, T with A, A with T, A with T, G with C. That gives G C A T T C.
5. C. 20 percent. A pairs with T, so A + T = 30 + 30 = 60 percent, and C and G share the other 40 percent equally, so G is 20 percent. If your child chose B, they gave all 40 percent to G and forgot that C takes half.
6. A gene codes for a protein, which is a chain of amino acids. 300 bases ÷ 3 bases per amino acid = 100 amino acids.
7. A. Two cells, each with the same chromosomes as the original. Mitosis copies the chromosomes first and then splits them, so each new cell is a copy of the original. If your child chose B, they mixed it up with the division that makes sex cells, which halves the chromosomes.
8. The number doubles each round: 1, then 2, then 4, then 8, then 16. So 2 × 2 × 2 × 2 = 16 cells.
9. Growth: more cells make a palm or a child bigger. Repair: new cells replace worn-out or damaged ones, like skin healing after a scrape.
10. D. Glucose + oxygen makes carbon dioxide + water + energy released. Respiration uses up glucose and oxygen and gives out carbon dioxide, water and energy. If your child chose A, they picked photosynthesis, which runs the other way.
11. A. One quarter. Each Tt parent gives T or t, so the offspring are TT, Tt, Tt and tt, and only tt is short: 1 out of 4. If your child chose C, they counted Tt as short, but T is dominant, so Tt is tall.
12. Sweat evaporates and takes heat from the skin, so the body cools to about 37 degrees Celsius. A rise in temperature triggers a response that lowers it, and this steady control is called homeostasis (negative feedback).

Section 2: Earth and space

1. D. Geosphere: the sand dunes of the desert. Sand dunes are sand and rock, which belong to the geosphere. If your child chose A, they put the Hajar mountains in the hydrosphere, but mountains are rock, so they are geosphere.
2. They take in carbon dioxide from the air (atmosphere) and water from the sea (hydrosphere), build living tissue (biosphere), and their roots hold the muddy soil in place (geosphere).
3. The Sun's heat makes sea and land water evaporate into vapour. The vapour rises, cools and condenses into clouds, then falls as rain and flows back to the sea.

4. B. Air over the land rises because land heats faster than water. Land heats faster than water, so air over land warms and rises, and cooler sea air moves in as the breeze. If your child chose A, they reversed it and thought the sea heats faster.
5. C. Summers are usually hot and humid near the coast. Climate is the usual pattern over many years, and usually hot and humid describes a pattern. If your child chose B, they picked one rainy evening, which is weather on a single day.
6. Added: burning fossil fuels and respiration by living things put carbon dioxide into the air. Removed: plants and algae take it in during photosynthesis.
7. Gases like carbon dioxide and methane absorb heat that Earth gives off and send part of it back down, which keeps the surface warm. Without this natural effect Earth would be too cold for most life.
8. A. It releases extra carbon dioxide, which traps more heat. Burning fuels releases extra carbon dioxide, which traps more heat in the air. If your child chose B, they thought the Sun gives off more heat, but burning does not change the Sun.
9. The panels make 5 × 30 = 150 kWh in the month. 150 ÷ 600 = 0.25, which is 25 percent.
10. Sunlight arrives every day and cannot be used up, so it is renewable. Coal took millions of years to form, so we use it far faster than it can be replaced.
11. B. The water becomes more acidic, so shells and skeletons are harder to build. Dissolved carbon dioxide makes seawater more acidic, which makes shells and skeletons harder to build. If your child chose C, they got the change backwards, because the water becomes more acidic, not less.
12. Melting ice uncovers dark water, which absorbs more sunlight than ice reflects. The extra heat melts more ice, so the change feeds itself (a positive feedback loop).

Section 3: Matter, forces and energy

1. A. Protons are positive and are found in the nucleus. Protons are positive and sit in the nucleus, with neutrons that have no charge. If your child chose B, they gave neutrons a positive charge, but neutrons are neutral.
2. Protons equal the atomic number, 11. A neutral atom has equal electrons, so 11. Neutrons = mass number - protons = 23 - 11 = 12.
3. C. 7. Valence electrons are the ones in the outer shell, and 2, 8, 7 ends with 7. If your child chose B (17), they counted all the electrons instead of only the outer shell.
4. Atoms in the same group have the same number of valence electrons, here one each. The outer electrons decide how an atom reacts, so they react alike.
5. Helium and neon have full outer shells, which is a stable arrangement. So their atoms have no need to gain, lose or share electrons.
6. D. Sodium gives an electron to chlorine, and the ions attract. Sodium gives one electron to chlorine, making Na+ and Cl- ions, and the opposite charges attract. If your child chose A, they described sharing electrons, which is a covalent bond, not the transfer in salt.
7. It is a covalent bond: atoms share electrons, usually between non-metal atoms, as in H2O. In an ionic bond one atom gives electrons to another, so the difference is sharing versus transferring.
8. C. 2 Mg + O2 → 2 MgO. Count atoms: 2 Mg + O2 gives 2 Mg and 2 O on the left, and 2 MgO gives 2 Mg and 2 O on the right. If your child chose D, they changed the formula to MgO2, but balancing only changes the numbers in front.

9. Put 2 in front of H_2O to make 4 H, then 2 in front of O_2 to make 4 O: $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$. Each side then has 1 C, 4 H and 4 O.
10. B. 40 g. Mass is conserved, so the products weigh the same as the reactants: 24 g + 16 g = 40 g. If your child chose A, they subtracted 24 - 16, but the reactants add together.
11. No. Iron joined with oxygen and water from the air to make rust, so the extra mass came from the oxygen. Atoms are rearranged, not created.
12. Left: C 6, H 12, O 12 + 6 = 18. Right: C 6, H 12, O 6 + 12 = 18, so it is balanced. A higher temperature or more reactant makes particles collide more often and harder, so the reaction goes faster.